

LAB 4 — VIBE CODING

Maze Chase | AI-assisted debugging and feature implementation

Student	Krishna G Pai
Student ID / SRN	Not provided in this conversation
Repository	https://github.com/krishna130506/11_maze_chase
Original repository	https://github.com/SETAPESU26/11_maze_chase
Language / Framework	Python / Pygame

Purpose

This report documents the AI-assisted workflow used for Lab 4 Vibe Coding. The assigned Maze Chase project was forked into the student's GitHub account, cloned locally, installed with its Python requirements, and run before code changes. Prompts were then used to fix existing bugs and implement the four README tasks incrementally. The coder's reported validation results and the implementation summaries shared during the conversation are recorded below.

Evidence note: This report is based on the prompts and coder-response screenshots provided in this chat. It records the validation the coder reported; it does not independently certify that every manual test was completed or that the final repository was pushed.

1. Repository setup and baseline

- Forked the assigned repository into the personal GitHub account: `krishna130506/11_maze_chase`.
- Cloned the fork into `C:\Users\krish\11_maze_chase` and opened it in VS Code.
- Confirmed Python 3.11.9 and installed dependencies with: `python -m pip install -r requirements.txt`.
- Ran the original game with: `python main.py`.
- Recorded the baseline gameplay video as `before_changes.mp4` (provided in the conversation).

2. AI-assisted bug fixes

Prompt used

Inspect my entire Maze Chase Python/Pygame project, including `main.py` and all files inside the game folder. First identify and fix the existing bugs without implementing new features yet. Preserve the current game design and controls. Explain each bug, its cause, and the files changed. Do not rewrite the entire project unnecessarily. Ensure the game still runs using `python main.py` and test the fixes where possible. After fixing the bugs, list the changes made and wait for my approval before implementing any new features.

Coder-reported changes

- Maze-wall collision: `Player_valid()` had checked grid bounds but ignored wall data. The coder updated `game/entities.py` so movement is blocked by closed horizontal and vertical edges while open passages remain traversable.
- HUD clipping: The instruction text and EXIT label did not fit their available space. The coder updated `game/game_engine.py` to use a smaller dedicated font and size/center the EXIT label.
- Reported validation: closed-wall and open-passage checks, text-fit checks, a headless game-loop smoke test, and launching `main.py` passed; no diagnostics were reported.

Commit recorded

7c98aba — Fix maze collision and HUD clipping. The terminal screenshot showed two files changed, 13 insertions and 3 deletions.

3. Task 1 — Three independently chasing enemies

Prompt used

Implement Task 1 from README.md in my Maze Chase project.
Add two additional enemies so there are three enemies in total. Each enemy must start at a different valid maze corner and independently chase the player using the existing BFS pathfinding implementation.
Reuse the existing enemy movement and pathfinding logic. Ensure all enemy spawn locations are valid and reachable.
Update movement, rendering, collision detection, and game-over logic for all three enemies. Preserve existing bug fixes.
Do not implement Tasks 2, 3, or 4 yet. Test the changes, explain manual verification, and do not commit or push.

Implementation summary

The coder reported three enemies spawning at distinct corners: top-right, bottom-right, and bottom-left. Each uses the existing `Enemy.update()` and BFS logic. The engine updates and draws all three and ends the game when any catches the player. The generated maze connects all cells and spawn reachability was checked.

Validation reported by the coder

Headless checks reportedly passed for distinct/reachable spawns, independent enemy movement, rendering, collision/game-over, and launching `main.py`. No diagnostics were reported.

Files reported as changed

`game/game_engine.py` and `README.md`

Planned / expected separate commit

Add multiple independently chasing enemies

Note: The screenshots in this conversation show the coder's implementation summaries and reported checks. They do not show terminal confirmation that this task's Git commit was created.

4. Task 2 — Progressive enemy difficulty

Prompt used

Implement Task 2 from README.md. Every 15 seconds of gameplay, reduce the enemy movement interval by 2, with a minimum of 5. Display the current speed tier in the HUD. Apply the change consistently to all three enemies, prevent duplicate changes within an interval, and reset timer/tier/intervals on restart. Preserve existing fixes and enemies. Implement Task 2 only, test, explain changed files and manual verification, and do not commit or push.

Implementation summary

The coder reported that the movement interval starts at 20 frames and decreases by 2 at each 15-second gameplay boundary, clamped at 5. The tier is derived from elapsed time; all three enemies receive the same interval. Restart resets the timer, tier, and intervals. The HUD displays the current speed tier and movement interval.

Validation reported by the coder

Simulated timing checks reportedly passed at tier boundaries, between boundaries, and at the minimum. Reset and synchronized intervals passed; all three enemies still chase independently, render, and main.py launches.

Files reported as changed

game/game_engine.py and README.md

Planned / expected separate commit

Add progressive enemy difficulty

Note: The screenshots in this conversation show the coder's implementation summaries and reported checks. They do not show terminal confirmation that this task's Git commit was created.

5. Task 3 — Power pellet and five-second freeze

Prompt used

Implement Task 3 from README.md. Add a yellow pellet at a valid maze location. When collected, freeze all three enemies for five seconds without freezing the player or game. Show a visible frozen indicator, resume automatically, prevent recollection until restart, use elapsed-time logic, reset pellet/freeze state on restart, preserve existing features, test, and do not commit or push. Implement Task 3 only.

Implementation summary

The coder reported a yellow pellet slightly left of maze center. Collecting it freezes all three enemies for five seconds using elapsed milliseconds, while player movement and the difficulty ramp continue. Frozen enemies receive a cyan outline and automatically resume chasing. The pellet cannot be collected again until restart; restart resets pellet and freeze state.

Validation reported by the coder

Deterministic checks reportedly passed for rendering/collection, all-enemy freeze, continued player/difficulty updates, recovery at five seconds, preventing recollection, resumed chase, and restart reset. main.py launched successfully with no diagnostics.

Files reported as changed

game/game_engine.py, game/entities.py, and README.md

Planned / expected separate commit

Add power pellet enemy freeze

Note: The screenshots in this conversation show the coder's implementation summaries and reported checks. They do not show terminal confirmation that this task's Git commit was created.

6. Task 4 — Survival score

Prompt used

Implement Task 4 from README.md. Increment survival score by 1 per active game update while the player is alive. Display "Survived: Xs", where X is score // 60. Stop increasing the score when the game ends, display the final score on the game-over overlay, reset on restart, preserve all existing features, test, explain changed files, and do not commit or push. Implement Task 4 only.

Implementation summary

The coder reported that the score increases once per active game update and stops on the frame the player wins or is caught. The HUD displays Survived: Xs using score // 60. Both end-game overlays show the final survival time, and restart resets the score. The existing two-row HUD and previous gameplay fixes were reported preserved.

Validation reported by the coder

The coder reported confirming that 60 active frames display 1s, the score stays unchanged after winning or being caught, overlays render, and restart resets the score. The main.py smoke test passed with no diagnostics.

Files reported as changed

game/game_engine.py and README.md

Planned / expected separate commit

Add survival score and game-over display

Note: The screenshots in this conversation show the coder's implementation summaries and reported checks. They do not show terminal confirmation that this task's Git commit was created.

7. Git workflow and commit history

The terminal screenshot confirms the bug-fix commit: 7c98aba Fix maze collision and HUD clipping. The subsequent task prompts explicitly instructed the coder not to commit or push. Therefore, the remaining four commits must be verified in the local repository before submission.

Order	Expected commit message	Status in available evidence
1	Fix maze collision and HUD clipping	Confirmed: 7c98aba
2	Add multiple independently chasing enemies	Needs local Git verification
3	Add progressive enemy difficulty	Needs local Git verification
4	Add power pellet enemy freeze	Needs local Git verification
5	Add survival score and game-over display	Needs local Git verification

Suggested verification commands

```
git status
git log -5 --oneline
git push origin main
```

Run git push origin main only after verifying the four feature commits, testing the final game, and ensuring all intended changes are committed.

8. Final manual test checklist

- Game launches with python main.py.
- Player cannot cross closed maze walls but can pass through open passages.
- HUD instructions and EXIT label fit correctly.
- Three enemies chase independently; collision with any enemy ends the game.
- Enemy interval decreases by 2 every 15 seconds, starts at 20, and never goes below 5.
- Yellow power pellet freezes all three enemies for five seconds; player and difficulty timer continue.
- Frozen indicator appears; enemies resume after five seconds.
- Survival counter increases once per active frame, stops at game end, and final time appears in the end-game overlay.
- Pressing R resets the game state, score, pellet, freeze state, and difficulty progression.
- Record the final gameplay video after all manual checks pass.

9. Submission deliverables

Place the following in the Lab-4 folder of the lab repository: (1) before_changes.mp4, (2) after_changes.mp4, (3) updated project source code, and (4) this AI conversation report as PDF or DOC. The baseline recording was uploaded in this conversation; the final recording and the final push are not confirmed by the available screenshots.

10. Conclusion

The Maze Chase project was developed incrementally with AI-assisted prompts. The reported changes address wall collision and HUD clipping, add three independently chasing enemies, progressive enemy difficulty, a five-second power-pellet freeze, and survival scoring. The coder reported successful targeted validation and smoke tests for each stage. Before final submission, manually verify the combined game, confirm all five separate commits, record the after video, export this report, and push the completed code to the personal fork.